

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – EXPERIMENTS

Course Code:	CSA0621	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Experiments (BL4, BL5)	Assessment:	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	V Deepa Dharshini	Reg. No.:	192525213
Section / Batch:	B tech AIML(2 nd year)	Date:	13-08-2026

Instructions to Students

- Write algorithm and execute code both experiments.
- Each questions 50 Marks. Total: 100 Marks.
- Follow a well-structured, systematic approach to design and implement an optimal solution.
- Provide insightful analysis with accurate validation, supported by clear documentation including diagrams, code, and results.

Question Type	Description	Questions	Total Marks
Experiments	Design and Code for Divide and Conquer Algorithms: Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication	Q1 – Q2	100

SECTION A – Experiments

**SIMATS Online Programming Lab**

V Deepa Dharshini
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CO3-AT1-Experiments

← Back

Language: Python C C++ Java

QUESTIONS

Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer
50 marks

2/2 answered

Q2 Divide and Conquer

50 marks

Evaluate the performance differences between Strassen's Matrix Multiplication and the standard matrix multiplication method by implementing both approaches. Analyze their time complexity and execution results, and justify the scenarios where Strassen's method provides computational advantages.

Your Code

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a, b, c, d;
6     int e, f, g, h;
7     int m1, m2, m3, m4, m5, m6, m7;
8     int c11, c12, c21, c22;
9
10    scanf("%d %d", &a, &b);
11    scanf("%d %d", &c, &d);
12
13    scanf("%d %d", &e, &f);
14    scanf("%d %d", &g, &h);
15
16    m1 = (a + d) * (e + h);
```

Input

```
1 2
3 4
5 6
7 8
```

Output

```
Strassen Result:
19 22
43 50
Time Complexity: O(n^2.807)
```

CO3-AT1-Experiments

← Back

Language: Python C C++ Java

QUESTIONS
Q1

Divide and Conquer
50 marks

Q2

Divide and Conquer
50 marks

2/2 answered

Q1 Divide and Conquer

50 marks

Evaluate the behavior of Quick Sort on nearly sorted datasets by implementing the algorithm and measuring execution time. Compare the results with random input datasets. Analyze how pivot selection impacts performance degradation. Interpret findings and justify techniques to improve efficiency.

Your Code

```
1 #include <stdio.h>
2 #include <time.h>
3
4 void quicksort(int a[], int low, int high)
5 {
6     int i, j, pivot, temp;
7
8     if (low >= high)
9         return;
10
11     i = low;
12     j = high;
13     pivot = a[(low + high) / 2];
14
15     while (i <= j)
16     {
```

Input

5
1 2 3 5 4

Output

Sorted array: 1 2 3 4 5
Execution time: 0.000000

Code of Conduct

I V Deepa Dharshini (192525213) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 
RUBRICS FOR CALCULATION

Experiments					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with	Organized documentation	Basic documentation	Poor or incomplete

		diagrams, code, and results	with necessary details	with missing details	documentation
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Marks Evaluation:

Experiments	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Marks (Each - 50)
Ex. 1						
Ex. 2						
Overall Total (100)						100 %

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Signature: _____	Signature: _____
Signature: _____		
Date: _____	Date: _____	Date: _____